

Sadia Sultana

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PROFILE SUMMARY

A Prospective Graduate Student for Fall 2027, completed B.Sc. in Computer Science and Engineering from the Islamic University of Technology (IUT), with research interests in Natural Language Processing, Large Language Models and Retrieval-Augmented Generation. Currently serving as a Lecturer in the Department of Software Engineering at Daffodil International University.

EDUCATION

B.Sc. in Computer Science and Engineering

Islamic University of Technology (IUT)

CGPA: **3.84**, First Class with Honours

June 2021 – October 2025

KB Bazar, Gazipur

RESEARCH EXPERIENCE

BanglaMedQA and BanglaMMedBench: Evaluating Retrieval-Augmented Generation Strategies for Bangla Biomedical Question Answering

2025

Accepted at **IEEE FMLDS 2026** [Preprint - [arXiv](#)]

Developed and evaluated multiple Retrieval-Augmented Generation (RAG) strategies for Bangla biomedical multiple choice question answering using two novel datasets - BanglaMedQA and BanglaMMedBench, comprising **2,000** questions. Evaluated retrieval and generation approaches using Large Language Models (LLMs), achieving the highest accuracy of **89.54%** with Agentic RAG. The study investigates knowledge-grounded generation for reliable biomedical question answering in Bangla.

WORK EXPERIENCE

Lecturer

Department of Software Engineering, Daffodil International University

January 2026 – Present

Birulia, Savar

- Teaching theory and laboratory undergraduate courses in Software Engineering Web Applications and Design Capstone Project.
- Served as Module Leader with responsibilities including coordinating course planning, learning outcomes, assessment, and course-related academic activities.

Flutter Developer Intern

Battery Low Interactive Ltd.

June 2024 – July 2024

Baridhara, Dhaka

- Completed a hands-on training program focused on building cross-platform mobile applications using Flutter and Dart.
- Gained experience in UI design, state management, API integration, Firebase services, and responsive layouts.

CERTIFICATIONS

The Daily Star Award for Edexcel High Achievers – A Level

Awarded for academic excellence in the Edexcel IAL examinations.

2020

The Daily Star Award for Edexcel High Achievers – O Level

Awarded for academic excellence in the Edexcel IGCSE examinations.

2018

TEST SCORES

IELTS Academic: 8.5

Listening: 9, Reading: 8.5, Writing: 7, Speaking: 8.5

EXTRA CURRICULAR ACTIVITIES

Assistant Publications Executive <i>IUT Computer Society</i> Co-editor of BEACON magazine for IUT ICT Fest 2024	April 2024
Volunteer <i>IUT Computer Society</i> Volunteered for IUT ICT Fest 2024 DevOps Hackathon	April 2024

RELEVANT UNDERGRADUATE COURSES

Data Structures and Algorithms | Structured Programming | Object-Oriented Programming | Database Systems | Artificial Intelligence | Machine Learning | Pattern Recognition | Web Applications | Operating Systems | Data Mining | Human Computer Interaction | Numerical Methods | Bioinformatics

TECHNICAL SKILLS

Languages: Python, C/C++, Dart, Java, SQL, HTML/CSS/JS
Backend Tech/Database: MySQL, Firebase, MongoDB
Deep Learning Frameworks (AI/ML): familiar with Pytorch, TensorFlow, Scikit-learn

PROJECTS

-  **MoodDecoder** | Pattern Recognition | Python
 - Built a Speech Emotion Recognition system using the MELD dataset and transformer-based architectures.
 - Performed preprocessing, fine-tuning, and evaluation to achieve high accuracy in emotion classification.
-  **DigitAid** | Machine Learning | Python
 - Developed a tool to aid dyslexic learners by recognizing handwritten digits through real-time prediction.
 - Implemented data capture, dataset generation, and model training modules to improve accessibility in education.
-  **PetPal** | Mobile App Development | Flutter, Dart
 - A comprehensive, one-stop solution for all pet-related needs, helping pet owners manage pet care and activities.
 - Includes features like - GPS tracking, community support, online veterinary care access, and online pet store.

REFERENCES

Tareque Mohmud Chowdhury Assistant Professor Department of Computer Science and Engineering Islamic University of Technology (IUT) Email: tareque@iut-dhaka.edu	Dr. Imran Mahmud Professor and Head of Department Department of Software Engineering Daffodil International University (DIU) Email: imranmahmud@daffodilvarsity.edu.bd
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